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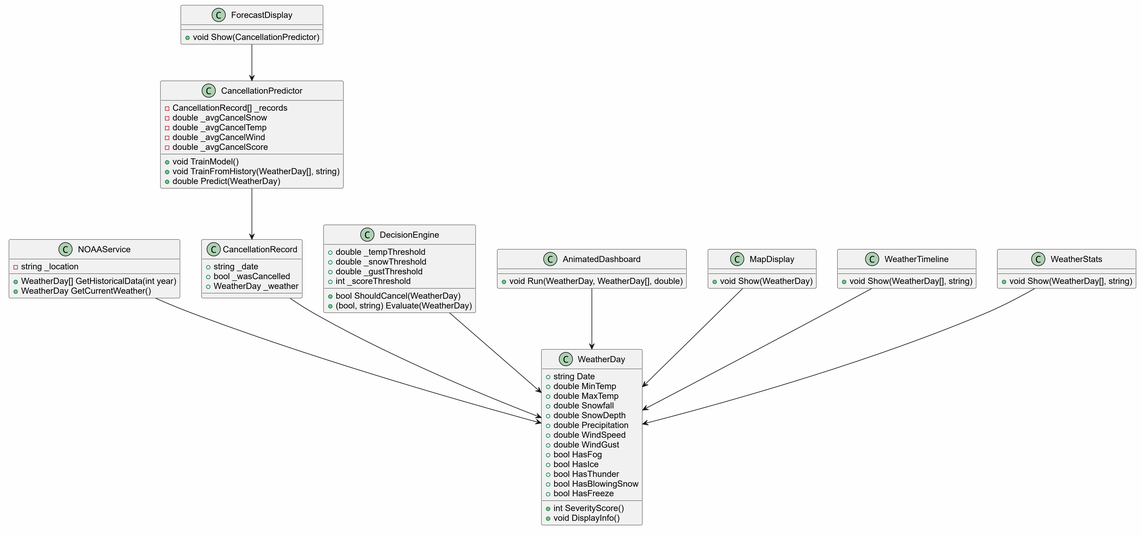
This editor runs entirely in your browser thanks to our JavaScript rendering engine. Edit the code on the left and see the diagram update instantly!

```
class CancellationPredictor {
  -CancellationRecord[] _records
  -double _avgCancelSnow
  -double _avgCancelTemp
  -double _avgCancelWind
  -double _avgCancelScore

  +void TrainModel()
  +void TrainFromHistory(WeatherDay[], string)
  +double Predict(WeatherDay)
}

class DecisionEngine {
  +double _tempThreshold
  +double _snowThreshold
  +double _gustThreshold
  +int _scoreThreshold

  +bool ShouldCancel(WeatherDay)
  +(bool, string) Evaluate(WeatherDay)
}

class AnimatedDashboard {
  +void Run(WeatherDay, WeatherDay[], double)
}

class MapDisplay {
  +void Show(WeatherDay)
}

class ForecastDisplay {
  +void Show(CancellationPredictor)
}

class WeatherTimeline {
  +void Show(WeatherDay[], string)
}

class WeatherStats {
  +void Show(WeatherDay[], string)
}

NOAAService --> WeatherDay
CancellationPredictor --> CancellationRecord
CancellationRecord --> WeatherDay
DecisionEngine --> WeatherDay
ForecastDisplay --> CancellationPredictor
AnimatedDashboard --> WeatherDay
MapDisplay --> WeatherDay
WeatherTimeline --> WeatherDay
WeatherStats --> WeatherDay

@enduml
```



🌱 PlantUML at a Glance

🚀 Getting Started

PlantUML is a highly versatile tool that facilitates the rapid and straightforward creation of a wide array of diagrams.

Utilizing a simple and intuitive language, users can effortlessly draft various types of diagrams. For a detailed exploration of the language's capabilities and syntax, please refer to the PlantUML Language Reference Guide.

If you are new to PlantUML, we recommend starting with our quick start page to get up and running quickly. Should you have any questions, our F.A.Q. page is a valuable resource. Additionally, PlantUML can be seamlessly integrated with a variety of other tools to enhance your workflow.

🧩 Supported UML Diagrams

With PlantUML, you can create well-structured UML diagrams, including but not limited to:

- Sequence diagram
- Usecase diagram
- Class diagram
- Object diagram
- Activity diagram *(Find the legacy syntax here)*
- Component diagram
- Deployment diagram
- State diagram
- Timing diagram

📈 Supported Non-UML Diagrams

Beyond the traditional UML diagrams, PlantUML also supports the creation of various other diagram types, such as:

- JSON (JavaScript Object Notation) data
- YAML (YAML Ain't Markup Language) data
- EBNF (Extended Backus-Naur Form) diagram
- Regex (Regular Expression) diagram
- Network diagram (nwdiag)
- Salt (Wireframe graphical interface or UI Mockups)
- Archimate diagram
- SDL (Specification and Description Language)
- Ditaa diagram
- Gantt diagram
- Chronology diagram
- MindMap diagram
- WBS (Work Breakdown Structure) diagram
- Mathematics with AsciiMath or JLaTeXMath notation
- Information Engineering (IE) diagram
- Entity Relationship (ER) diagram (Chen's notation)
- Chart diagram
- Files (files tree) diagram

📣 Additional Features

Enhance the dynamism and informativeness of your diagrams with these additional features:

- Hyperlinks and tooltips to provide extra context and interactivity
- Rich text formatting, emoticons, Unicode, and icons with Creole for a visually appealing presentation
- OpenIconic icons for enhanced visual representation
- Sprite icons to add custom symbols
- AsciiMath mathematical expressions for precise mathematical representation

📥 Input Formats

PlantUML allows you to generate diagrams from various source input formats:

- Source input data: *How and where diagrams can be written*

You can choose from different internal encodings:

- PlantUML Text Encoding

⚙️ Layout Engines and Options

PlantUML allows you to use several different layout engines:

- Graphviz is the default engine. Relies on an external program;
- Smetana *(Use `!pragma layout smetana` in the file or `-Playout=smetana` on the command-line.)* is a internal port of Graphviz to Java. Tends to make slightly straighter arrows;
- VizJs uses JavaScript. *(Use `-graphvizdot vizjs` on the command line.)* Tends to space out nodes more, resulting in a bigger diagram;
- ELK (Eclipse Layout Kernel). *(Use `!pragma layout elk` in the file or `-Playout=elk` on the command-line.)* Supports only orthogonal layout, and doesn't cover all features.

Sequence diagrams have 2 engines:

- *Puma* is the older and still default engine;
- Teoz *(Use `!pragma teoz true` in the file or `-Pteoz=true` on the command-line)* is a new engine that has some unique features: *anchors (named arrows/messages), duration, nested boxes...*

  Achieving good layout with PlantUML is sometimes non-trivial. *See PlantUML GraphViz Layout by Mark George at University of Otago for more advice.*

📥 Output Formats

Export your diagrams in a variety of formats to suit your needs, including:

- PNG for easy image sharing
- SVG for scalable vector graphics
- LaTeX for high-quality typesetting
- EPS (Encapsulated PostScript) for used with LaTeX
- ASCII art *(available only for sequence diagrams)* for a text-based representation

🎉 Enjoy

Get started today and elevate your diagramming capabilities with **PlantUML**. Transform your ideas into clear and professional diagrams effortlessly 🎉 !